

Franck-Hertz Phys 20 CL Lab

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May 2026

1 Abstract

The focus of this lab was to determine the first excitation energy of mercury vapor. We imitated the famous Franck-Hertz experiment by accelerating thermionically emitted electrons through mercury vapor and measuring the collection current as a function of accelerating voltage. The current showed repeated peaks and troughs, indicating discrete energy losses from inelastic electron-mercury collisions. From the spacing between successive current maxima, the first excitation energy was found to be 4.86 ± 0.12 eV, which is close to the accepted spectroscopic value of 4.89 eV.

2 Apparatus

- filament cathode
- wire mesh anode
- collection electrode
- mercury (Hg)
- oven
- large variac
- small variac
- potentiometers
- DC power supply

3 Theory & Procedure

The original experiment was not designed to confirm quantum theory, but it incidentally provided strong evidence that atoms accept energy in discrete levels. In our version of the Franck-Hertz experiment, electrons were emitted by a

heated filament cathode through thermionic emission and accelerated towards an anode grid by an accelerating voltage V_a . Some of these electrons had sufficient energy, which allowed them to pass through the grid onto the collecting electrode, where the current was measured as the accelerating voltage varied. A large variac was used to heat the oven until it settled at 190° C, vaporizing mercury to provide the possibility of collisions between electrons and mercury atoms between the filament cathode and the anode grid. We heated the filament cathode using a small variac and used the potentiometers to adjust the accelerating and retarding voltages, the accelerating voltage powered by the DC power supply. The retarding voltage is imposed to filter electrons with sufficient energies and is fixed in the experiment. It is powered by a 1.5 volt battery, and an electrometer is used to measure the current at the collecting electrode.

4 Data

The voltage measurements on the digital multimeters were displayed with a resolution of 0.01 V, and similarly the resolution of the current was 0.01 nA. Therefore, the experimental uncertainty in each measurement is ± 0.005 nA for the collection current and ± 0.005 V for the accelerating voltage. To calculate the first excitation energy of mercury, we use the fact that an electron accelerated through a potential difference V gains kinetic energy eV . Taking into account the shift in a graph of I vs V due to the presence of a contact potential between the filament cathode and the anode grid, we take the difference between subsequent maxima to find the average first excitation energy. To obtain an uncertainty in the mean first excitation energy, I implemented the formula

$$U_m = \frac{s}{\sqrt{N}} \quad (1)$$

where s is the standard deviation of measured peak spacings, and N is the number of spacings used in the average. The standard deviation was calculated using the NumPy standard deviation function.

5 Analysis & Conclusion

The local maxima of the current curve were identified using Python, and the differences between successive peak voltages were calculated. These voltage spacings were then averaged to determine the first excitation potential. As seen in Figure 1, our value of 4.86 ± 0.12 eV was close to the spectroscopically measured value of 4.89 eV, with a percentage error of just 0.61%. Overall, this was a fairly successful measurement of E_1^{Hg} , and the small discrepancy likely arose from fluctuations in the device readings and uncertainty in finding the peaks.

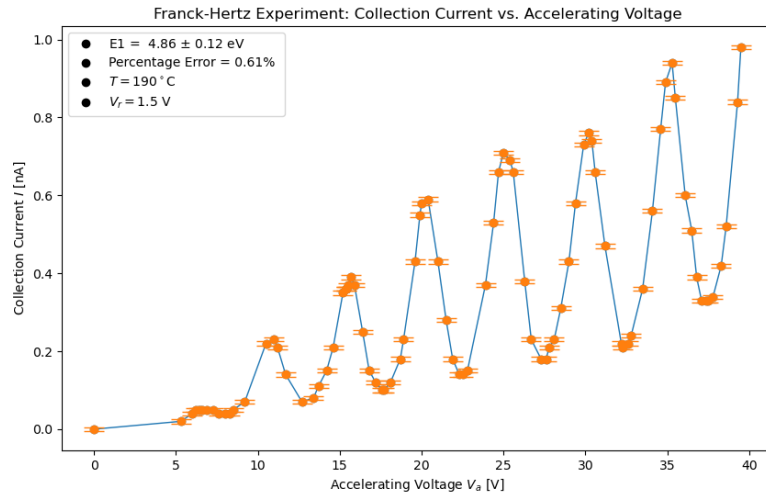


Figure 1: Collection current as a function of accelerating voltage for the Franck-Hertz experiment. The retarding potential was fixed at $V_r = 1.5 \text{ V}$, and the oven temperature was 190°C . Error bars represent the digital readout uncertainties.